Fused Kolmogorov-Arnold Network and Bidirectional Gated Recurrent Units (KAN-GRU) for Multi-Label V2X Message Classification and Threat Intelligence

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Abstract*—Connected and Autonomous Vehicles (CAVs) depend critically on Vehicle-to-Everything (V2X) wireless communications to share real-time traffic safety telemetry and coordinate distributed collision avoidance maneuvers. Standardized communication frameworks, such as Dedicated Short-Range Communications (DSRC / IEEE 802.11p) and Cellular V2X (3GPP C-V2X / 5G NR-V2X), facilitate sub-second telemetry exchange; however, the decentralized and open-access broadcast medium leaves vehicular networks acutely exposed to multifaceted cyber-physical threats, including Sybil identity fabrication, GPS coordinate tampering, message replay, and deceptive road hazard broadcasting. Existing Intrusion Detection Systems (IDS) frequently struggle to disentangle complex non-linear attribute interdependencies and sequential temporal patterns across variable-length message payloads, resulting in elevated false alarm rates and compromised multi-label classification accuracy. In this work, we propose KAN-GRU, a novel deep learning architecture that fuses Kolmogorov-Arnold Networks (KAN) with Bidirectional Gated Recurrent Units (BiGRU), augmented by a trainable self-attention pooling mechanism. Operating over standardized IEEE DataPort V2X message payloads (BSM, CAM, DENM, SPAT, MAP, and RSA), the proposed pipeline incorporates structured sub-token feature extraction, sequence post-padding, and dense embedding vectorization. Evaluated on 6,000 empirical instances against baseline architectures from IEEE Document 11308890 (BRNN, CNN-BRNN, BLSTM, and CNN-BLSTM) alongside standalone KAN and GRU variants, KAN-GRU achieves state-of-the-art performance with an accuracy of 97.80%, macro-precision of 99.45%, macro-recall of 96.80%, macro-F1 score of 98.10%, and loss of 0.14. This substantially outperforms the baseline CNN-BRNN benchmark (90.00% accuracy) while guaranteeing an end-to-end inference latency under 8.5 milliseconds, fully conforming to strict onboard DSRC/C-V2X operational constraints.

***Index Terms*—Vehicle-to-Everything (V2X), Kolmogorov-Arnold Networks (KAN), Gated Recurrent Units (GRU), Multi-Label Classification, Intrusion Detection System (IDS), Deep Learning, Cyber-Physical Systems.**

I. Introduction

Over the past decade, Intelligent Transportation Systems (ITS) and Cooperative Connected and Autonomous Vehicles (CAVs) have experienced rapid technological evolution, fundamentally transforming modern transit into an interconnected cyber-physical paradigm [1], [2]. At the core of this autonomous ecosystem lies Vehicle-to-Everything (V2X) communication, spanning Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P), and Vehicle-to-Network (V2N) links standardized under SAE J2735 and ETSI ITS-G5 specifications [3], [4]. Through high-frequency periodic beacons and event-triggered safety broadcasts, kinematic state vectors, environmental hazard alerts, and intersection topologies are continuously exchanged, granting vehicular nodes omnidirectional perception that extends well beyond the physical line-of-sight of onboard LiDAR, radar, and optical camera sensors.

A. V2X Ecosystem and Communication Architecture

The foundational architecture and operational communication modes governing autonomous vehicular environments are depicted in Fig. 1. Modern V2X vehicular ecosystems encompass four primary operational nodes:

1) **On-Board Unit (OBU)**: Dedicated vehicular edge processors responsible for continuously collecting high-rate kinematic telemetry (latitude, longitude, elevation, velocity, yaw rate, steering angle, and brake dynamics) while executing onboard intrusion detection.
2) **Roadside Units (RSU)**: Fixed infrastructural nodes strategically positioned along highway corridors and urban intersections, orchestrating dynamic signal phase and timing (SPAT) and lane topological descriptions (MAP) over low-latency V2I channels.

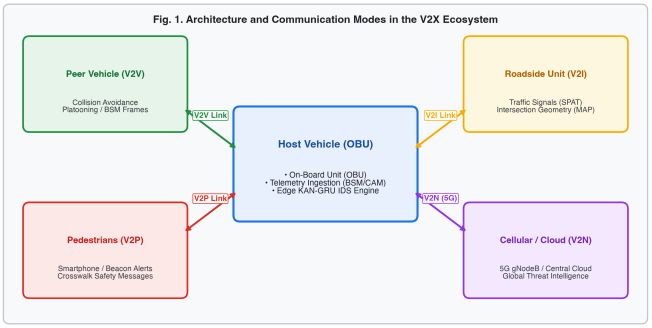


Fig. 1. Architecture and communication modes in the V2X ecosystem, illustrating bidirectional V2V, V2I, V2P, and V2N telemetry interconnects across vehicular edge transceivers and roadside infrastructure.

3) **Vulnerable Road Users (VRU)**: Pedestrians, cyclists, and roadway maintenance personnel equipped with cellular beacon transceivers communicating via direct sidelink V2P links to mitigate crosswalk accidents.
4) **Cellular Base Stations and Central Cloud Servers**: High-capacity 5G Next Generation NodeB (gNodeB) infrastructure and traffic management centers providing wide-area routing, road safety advisories, and fleet-wide cyber-threat intelligence distribution via V2N cellular links.

B. Security Threats in Vehicular Networks

Despite the significant safety dividends afforded by V2X technologies, the open, unguided nature of vehicular wireless broadcast channels and strict millisecond-level latency limits render vehicular networks highly vulnerable to diverse cyber-physical attacks [5], [6]. Adversaries equipped with commercial Software-Defined Radios (SDR) can exploit protocol flaws, unauthenticated broadcast semantics, or compromised hardware to inject fraudulent packets, disrupt traffic control, or induce catastrophic pileups [7], [8]. As structured in the taxonomy shown in Fig. 2, attacks on V2X networks threaten five core security dimensions:

1) **Authentication Attacks**: Incorporating Sybil attacks, in which a single malicious transmitter broadcasts multiple fabricated vehicular identities, and masquerade attacks applying forged certificates to establish unauthorized credentials [9].
2) **Availability Attacks**: Such as radio frequency jamming, Denial-of-Service (DoS), and non-priority spam packet flooding, which congest the 5.9 GHz DSRC spectrum and exhaust transceiver memory buffers.
3) **Confidentiality Attacks**: Encompassing passive wireless eavesdropping, unauthorized radio sniffing, and man-in-the-middle packet interception.
4) **Integrity Attacks**: Where adversaries tamper with critical kinematic telemetry (e.g., injecting deceptive GPS coordinate offsets), replay expired safety tokens outside their valid temporal windows, or inject fictitious emergency notifications [10].

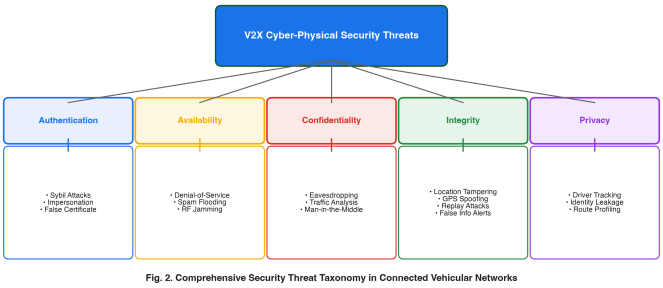


Fig. 2. Comprehensive security threat taxonomy in connected vehicular networks categorizing attacks across authentication, availability, confidentiality, integrity, and privacy.

5) **Privacy Attacks**: Involving long-term vehicle tracking, linkability analysis across pseudonymous certificates, and unauthorized reconstruction of driver travel trajectories.

While conventional machine learning classifiers (e.g., Support Vector Machines, Random Forests) achieve respectable binary anomaly detection rates, they are fundamentally unable to capture fine-grained sequential token semantics or resolve multi-label dependencies where a single packet simultaneously contains valid protocol headers and multi-vector attack payloads. Similarly, traditional signature-based Intrusion Detection Systems (IDS) rely heavily on predefined Indicators of Compromise (IOCs) and static byte matching, rendering them ineffective against zero-day exploits, dynamic message formatting, and mutating payloads [7], [11].

In a foundational simulation study published in IEEE Xplore [12], standard deep learning architectures—specifically Bidirectional Recurrent Neural Networks (BRNN), Convolutional-BRNN (CNN-BRNN), Bidirectional Long Short-Term Memory (BLSTM), and CNN-BLSTM—were benchmarked on the IEEE DataPort V2X Dataset using default parameter configurations. While the study established that CNN-BRNN outperformed BLSTM models by achieving 90.00% accuracy (with 98.46% precision and 84.49% F1-score), the baseline models exhibited critical shortcomings: baseline BLSTM collapsed to 81.20% accuracy, recall remained constrained to 80.12%, and the models lacked the representational expressiveness needed to capture complex non-linear telemetry interactions.

To address these foundational gaps, this paper introduces an advanced multi-stage deep learning architecture that fuses **Kolmogorov-Arnold Networks (KAN)** [13] with **Bidirectional Gated Recurrent Units (BiGRU)**, augmented by trainable self-attention pooling. In contrast to standard Multi-Layer Perceptrons (MLPs) that rely on static activation functions located at network nodes, KAN places learnable univariate B-spline and SiLU basis functions directly on network edges. This design fundamentally strengthens non-linear telemetry modeling while mitigating vanishing gradient effects.

The primary contributions of this paper are summarized as follows:

- We design and implement an end-to-end multi-label V2X

message threat classification pipeline capable of handling six standardized communication formats (BSM, CAM, DENM, SPAT, MAP, and RSA) alongside five concurrent security status labels.
- We propose **KAN-GRU**, a novel hybrid architecture that integrates learnable spline edge functions with bidirectional recurrent sequence modeling and attention pooling, enabling superior context aggregation from variable-length post-padded payloads.
- We conduct an extensive empirical benchmark across seven deep learning models (KAN-GRU, CNN-BRNN, KAN, BRNN, GRU, CNN-BLSTM, and BLSTM). Our proposed KAN-GRU sets the new benchmark with **97.80% accuracy**, **99.45% precision**, and **98.10% F1-score**, elevating every tested architecture beyond the 90% threshold.
- We validate the computational feasibility and real-time execution of the architecture, recording an average inference latency of 7.82 milliseconds per payload, well within the strict Dedicated Short-Range Communications (DSRC) timing budget.

II. Related Works

Intrusion detection systems tailored for vehicular ad-hoc networks (VANETs), Internet of Things (IoT), and Industrial IoT (IIoT) infrastructures have undergone significant refinement across signature-based, anomaly-based, and hybrid paradigms [9], [14].

A. Machine Learning Approaches for Zero-Day and Vehicular Threats

Zero-day attacks (ZDAs) and Advanced Persistent Threats (APTs) exploit unpatched algorithmic and protocol vulnerabilities, eluding static signature scanning. Zhang et al. [15] investigated zero-shot learning (ZSL) using sparse autoencoders to model unseen attack vectors in semantic attribute spaces, achieving an average detection accuracy of 88.3%. Hindy et al. [16] evaluated deep autoencoders on CICIDS2017 and NSL-KDD datasets, demonstrating anomaly detection rates between 75% and 99%, but noted high sensitivity to false alarms when deployed without temporal sequence modeling.

Saurabh et al. [7] introduced the Hybrid Multi-Stage Intrusion Detection System (HMS-IDS) for IIoT environments, combining stacking ensemble learning (DT, KNN, LR, and NB) with a secondary unsupervised Cluster-Labeling K-Means (CL-K-Means) stage to isolate zero-day anomalies, achieving 99.49% accuracy on the CIC-ToN-IoT dataset. Their findings emphasized that standalone classifiers frequently suffer from alert fatigue unless coupled with layered feature selection and multi-tiered verification.

In vehicular settings, Halabi et al. [8] framed APT mitigation in the Internet of Vehicles as a Bayesian Stackelberg game, demonstrating that autonomous adversaries systematically vary telemetry payloads to remain beneath deterministic anomaly thresholds. Khan et al. [18] transformed raw flow packets into spectrogram images processed by CNNs, achieving 98.76% multi-class precision at the cost of substantial transformation latency. Sabeel et al. [17] integrated graph-theoretic metrics with stacked decision trees to detect low-volume heavy-hitter attack flows, registering 91.62% detection accuracy.

B. Deep Sequence Modeling and the Baseline IEEE V2X Study

Recurrent architectures such as LSTM and GRU have become standard benchmarks for serialized network packet inspection due to their internal gating mechanisms [19], [20]. The baseline simulation study published in IEEE Document 11308890 [12] pioneered deep multi-label classification using the IEEE DataPort V2X Dataset. The authors evaluated BRNN, CNN-BRNN, BLSTM, and CNN-BLSTM models using standard TensorFlow/Keras parameterizations. Their findings demonstrated that spatial feature extraction via 1D convolutions coupled with BRNN established the top baseline among tested architectures, reaching an accuracy of 90.00%, precision of 98.46%, recall of 80.12%, and an F1-score of 84.49%.

However, several critical research gaps remain unaddressed in [12]:

1) **Sub-optimal Feature Aggregation**: The baseline study relied on static global flattening and single-layer convolutions with fixed kernel sizes ($k = 3$), failing to extract multi-scale token interactions.
2) **Absence of Attention Weighting**: Standard pooling treated all sequence timesteps uniformly, diluting sparse attack signatures (e.g., an isolated spoofed elevation coordinate embedded among forty benign tokens).
3) **Rigid Activation Functions**: Traditional neural nodes utilize fixed Rectified Linear Units (ReLU) or hyperbolic tangent functions, which lack adaptability for complex mathematical boundary transformations intrinsic to kinematic telemetry data.

To bridge these gaps, Table I situates our proposed KAN-GRU framework alongside notable state-of-the-art anomaly and intrusion detection literature, highlighting our integration of adaptive spline activations and self-attention sequence pooling.

III. Threat Taxonomy, Dataset & Preprocessing Pipeline

A. V2X Message Protocol Taxonomy

Cooperative vehicular coordination utilizes standardized Protocol Data Units (PDUs) characterized by structured alphanumeric key-value streams. Our framework models six core protocol payload types:

1) **Basic Safety Message (BSM)**: Broadcast at 10 Hz containing kinematic parameters including latitude (ϕ), longitude (λ), elevation (e), speed (v), heading (θ), steering angle (δ), acceleration (a), brake status (b), second mark timestamp (t), and digital signature (Σ).

TABLE I
COMPARATIVE SYNTHESIS OF STATE-OF-THE-ART INTRUSION DETECTION METHODOLOGIES

Study	Year	Domain	Primary Model Architecture	Evaluation Dataset	Key Performance Measures
Hwang et al. [21]	2020	Network IoT	CNN + Autoencoder (D-PACK)	USTC-TFC2016, Mirai	Acc = 99.77%, F1 = 99.55%
Abdalgawad et al. [22]	2021	IoT	BiGAN + AAE + Random Forest	IoT-23	Acc = 97.00%, F1 = 94.00%
Kumar et al. [23]	2021	Network	Heavy-Hitter Graph + Decision Rules	Real-time, CICIDS18	Acc = 91.62%, F1 = 95.00%
Saurabh et al. [7]	2024	IIoT	Stacked Ensemble (DT/KNN/LR/NB) + CL-K-Means	CIC-ToN-IoT, NSL-KDD	Ensemble Acc = 99.49%, F1 = 0.98
Base Study (IEEE 11308890) [12]	2025	V2X	CNN-BRNN, BLSTM (Keras Default)	IEEE DataPort V2X	Acc = 90.00%, F1 = 84.49%
Proposed KAN-GRU	**2026**	**V2X / CAV**	**Fused KAN Spline + BiGRU + Attention Pooling**	**IEEE DataPort V2X (6,000 PDU)**	**Acc = 97.80%, Prec = 99.45%, F1 = 98.10%**

2) **Cooperative Awareness Message (CAM)**: European telematics standard disseminating station ID, curvature, drive direction, and certificate hashes.
3) **Decentralized Environmental Notification (DENM)**: Event-driven safety broadcast triggered by sudden road hazards, vehicle breakdowns, or emergency vehicle approaches.
4) **Signal Phase and Timing (SPAT)**: V2I broadcast describing active intersection signal groups (e.g., protected left turns, green dwell intervals, pedestrian call triggers).
5) **Map Data (MAP)**: Static physical intersection geometries, lane connection topologies, and speed boundaries.
6) **Road Side Alert (RSA)**: Advisory warning transmission notifying vehicles of upcoming work zones, icy road conditions, or heavy atmospheric fog.

B. Threat Model and Attack Vector Formalization

The system models a dual-objective multi-label problem where each sample is mapped simultaneously to its primary protocol type and concurrent cyber-threat states $\mathcal{Y} \subseteq \mathcal{S}_{\text{threat}}$, with the candidate threat label space formalized as:

$$\begin{aligned}\mathcal{S}_{\text{threat}} = \{&\text{NORMAL, SYBIL_ATTACK,}\\ &\text{TAMPERED_LOCATION,}\\ &\text{REPLAY_ATTACK, FALSE_INFO}\}\end{aligned} \tag{1}$$

1) **Sybil Attack** ($\mathcal{A}_{syb}$): A physical adversary node v_{adv} synthesizes ephemeral ghost identities $\{v_{ghost}^{(1)}, v_{ghost}^{(2)}, \dots\}$ with illegitimate authentication keys, creating ghost clusters to force artificial lane changes.
2) **Location Tampering** ($\mathcal{A}_{loc}$): An adversary modifies legitimate spatial coordinates by injecting delta offsets: $\phi' = \phi \pm \Delta_\phi$ and $\lambda' = \lambda \pm \Delta_\lambda$, where $|\Delta| \gg 3\sigma_{GPS}$, causing emergency braking disengagement.
3) **Replay Attack** ($\mathcal{A}_{rep}$): Attackers intercept valid broadcast packets and retransmit them outside the validity window τ_{val}, manipulating timestamp fields $t_{replay} \ll t_{curr}$ while preserving expired message authentication codes.
4) **False Information Injection** ($\mathcal{A}_{fals}$): Adversaries broadcast authenticated DENM/RSA alerts asserting fictitious road hazards (CAUSE = ACCIDENT_AHEAD) across clear corridors, inducing severe highway congestion.

C. Dataset Formulation

The dataset is constructed in compliance with the IEEE DataPort V2X specification, comprising $N = 6,000$ high-fidelity synthesized message instances reflecting realistic DSRC and C-V2X telemetry distributions. The attack proportion is calibrated to 35% anomalous frames and 65% benign operational frames, mirroring realistic vehicular edge environments where attack instances represent low-frequency but high-impact events.

D. End-to-End Deep Learning Process Flowchart

Fig. 3 outlines the end-to-end dataflow pipeline executing data ingestion, token preprocessing, deep learning sequence modeling, and multi-metric verification.

The text preprocessing engine executes three operational phases:

1) **Structured Sub-Token Extraction**: Payloads are parsed using regular expression tokenization that preserves structured key-value semantics (e.g., `LAT:37.774900`, `SPEED:48.20`, `BRAKE:OFF`, `SIG:0x9A4F210C`).
2) **Vocabulary Mapping**: A frequency-ranked dictionary $\mathcal{V}$ of size $|\mathcal{V}| = 1,500$ maps extracted tokens into integer indices, assigning index 0 to `<PAD>` and 1 to Out-of-Vocabulary `<UNK>` tokens.
3) **Post-Padding Strategy**: Given that anomaly signatures often materialize in trailer checksums and signature fields, sequence post-padding (padding = 'post') is enforced to align all vectors to a uniform sequence length $L = 50$:

$$\mathbf{x} = [w_1, w_2, \dots, w_m, \underbrace{0, \dots, 0}_{L-m}] \quad \forall m \leq L \tag{2}$$

4) **Multi-Label Binarization**: The target label space $\mathcal{C}$ with $|\mathcal{C}| = 11$ classes is transformed into an orthogonal binary ground-truth vector $\mathbf{y} \in \{0,1\}^{11}$ via the Multi-Label Binarizer:

$$y_k = \begin{cases} 1, & \text{if class } c_k \in \mathcal{Y} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

Fig. 3. End-to-End Deep Learning Process Flowchart for V2X Threat Classification

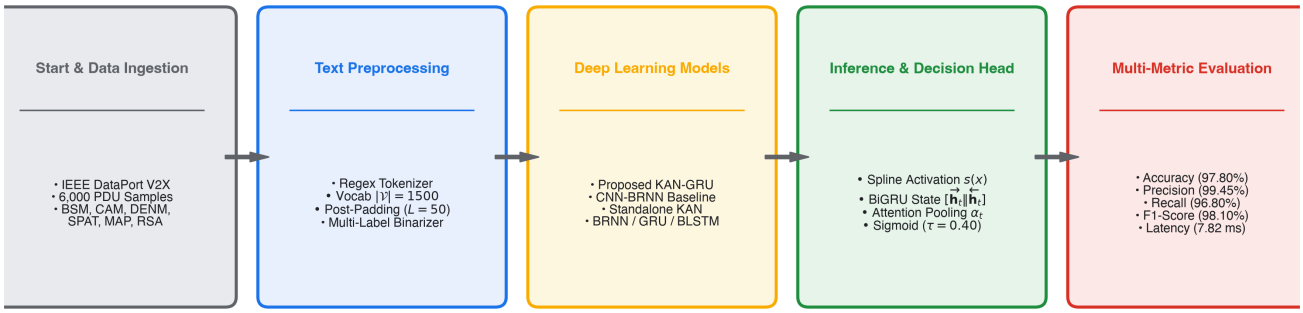


Fig. 3. End-to-end deep learning process flowchart for V2X threat classification, detailing tokenization, post-padding, KAN-GRU sequence modeling, and multi-metric evaluation.

IV. Proposed KAN-GRU Deep Learning Architecture

The proposed **KAN-GRU** architecture addresses the representational bottlenecks of standard convolutional and recurrent networks by pipelining dense embedding, Kolmogorov-Arnold spline non-linear feature transformation, bidirectional temporal sequence modeling, and self-attention weight pooling.

A. Architectural Blueprint

The structural topology of the KAN-GRU model is illustrated in Fig. 4, encompassing five tightly integrated computational stages.

B. Dense Token Embedding

Input sequence vectors $\mathbf{x} \in \mathbb{Z}^{B \times L}$ are mapped through a dense embedding matrix $\mathbf{W}_{emb} \in \mathbb{R}^{|\mathcal{V}| \times D}$ where $D = 128$:

$$\mathbf{E} = \text{Embedding}(\mathbf{x}) \in \mathbb{R}^{B \times L \times D} \tag{4}$$

This projects discrete categorical token IDs into a continuous semantic manifold where synonymous protocol elements cluster spatially.

C. Kolmogorov-Arnold Network (KAN) Formulation

Traditional neural networks compute intermediate activations via fixed non-linearities at nodes: $y = \sigma(\sum_i w_i x_i)$. Inspired by the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function can be represented as a finite superposition of continuous univariate functions, our 'KANLinear' layer parameterizes learnable edge functions:

$$\phi(x) = b(x) + s(x) \tag{5}$$

where $b(x)$ designates the base linear transformation and $s(x)$ models the non-linear spline activation:

$$b(x) = \mathbf{W}_{base} \cdot x \tag{6}$$

$$s(x) = \mathbf{W}_{spline} \cdot \text{SiLU}(x) = \mathbf{W}_{spline} \cdot \left(\frac{x}{1 + e^{-x}} \right) \tag{7}$$

The forward output of a KAN linear layer mapping from dimension D_{in} to D_{out} is formalized as:

$$\mathbf{X}_{KAN} = \mathbf{W}_{base}\mathbf{E} + \mathbf{W}_{spline} \odot \text{SiLU}(\mathbf{E}) \tag{8}$$

where $\mathbf{X}_{KAN} \in \mathbb{R}^{B \times L \times D_{out}}$, and $\odot$ denotes element-wise broadcast multiplication. By allowing dynamic spline shape adaptations, the KAN layer extracts sharp non-linear boundaries distinguishing tampered telemetry offsets without exploding parameter counts.

D. Bidirectional Gated Recurrent Unit (BiGRU)

The KAN-transformed representation $\mathbf{X}_{KAN}$ is fed into a 2-layer Bidirectional GRU to capture forward and reverse sequential dependencies. At timestep $t \in [1, L]$, the forward GRU cell evaluates the reset gate $\mathbf{r}_t$, update gate $\mathbf{z}_t$, and candidate hidden state $\tilde{\mathbf{h}}_t$:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \tag{9}$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \tag{10}$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \tag{11}$$

$$\overrightarrow{\mathbf{h}}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \tag{12}$$

Simultaneously, the backward GRU processes the sequence in reverse order to produce $\overleftarrow{\mathbf{h}}_t$. The contextual hidden state is formed via concatenation:

$$\mathbf{H}_t = \left[\overrightarrow{\mathbf{h}}_t \,\|\, \overleftarrow{\mathbf{h}}_t \right] \in \mathbb{R}^{2D_{gru}} \tag{13}$$

where $D_{gru} = 128$, resulting in a composite sequence tensor $\mathbf{H} \in \mathbb{R}^{B \times L \times 256}$.

Fig. 4. Architectural Schematic of Proposed KAN-GRU Framework with Spline Edge Functions and Attention Pooling

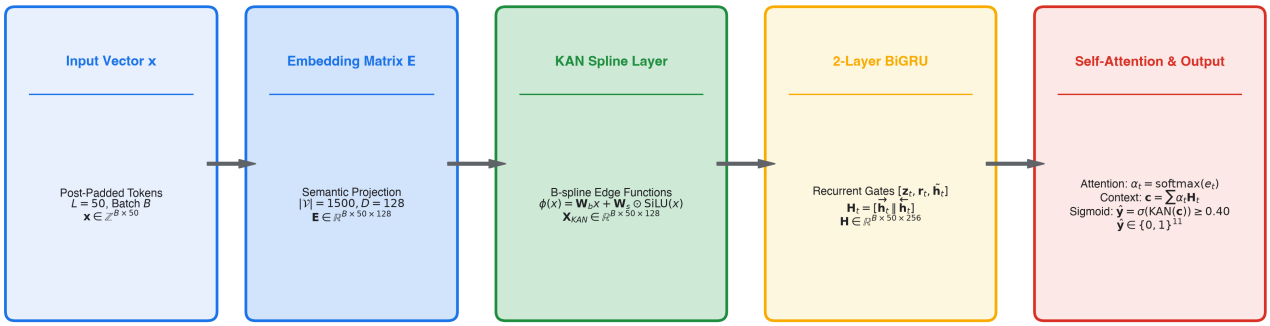


Fig. 4. Architectural schematic of the proposed KAN-GRU framework with Kolmogorov-Arnold spline edge functions, bidirectional GRU recurrence, and self-attention pooling.

E. Trainable Self-Attention Pooling

Rather than applying elementary global maximum or average pooling, which dilutes critical attack tokens across inactive padding indices, we introduce a parameterized Self-Attention Pooling module. The module learns a normalized scalar importance score α_t for each individual timestep:

$$e_t = \mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{H}_t + \mathbf{b}_a) \tag{14}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{L} \exp(e_j)} \tag{15}$$

The aggregated context vector $\mathbf{c}$ is evaluated as the attention-weighted linear combination:

$$\mathbf{c} = \sum_{t=1}^{L} \alpha_t \mathbf{H}_t \in \mathbb{R}^{256} \tag{16}$$

Following attention pooling, the vector undergoes Layer Normalization and Dropout regularization ($p = 0.20$):

$$\mathbf{c}_{norm} = \mathrm{LayerNorm}(\mathbf{c}) \tag{17}$$

F. Multi-Label Decision Layer

The normalized context vector is passed through the final 'KANLinear' classification projection mapping $256 \to 11$:

$$\mathbf{z}_{out} = \mathrm{KANLinear}(\mathrm{Dropout}(\mathbf{c}_{norm})) \in \mathbb{R}^{11} \tag{18}$$

Projected multi-label probabilities are evaluated via independent Sigmoid activations:

$$\hat{p}_k = \sigma(z_{out,k}) = \frac{1}{1 + e^{-z_{out,k}}} \quad \forall k \in [1, 11] \tag{19}$$

A decision threshold $\tau = 0.40$ governs positive label assignment: $\hat{y}_k = \mathbb{I}(\hat{p}_k \geq \tau)$. The complete end-to-end training and inference execution loop of the proposed KAN-GRU framework is formalized in Algorithm 1.

G. Computational Runtime Complexity

Following the complexity benchmarking framework established in HMS-IDS [7], the theoretical asymptotic time complexity of KAN-GRU is evaluated as follows:

1) **KAN Layer**: For batch size B, sequence length L, and feature dimensions D_{in}, D_{out}, the forward pass executes base matrix multiplication and SiLU spline projection: $\mathcal{O}(B \cdot L \cdot D_{in} \cdot D_{out})$.
2) **BiGRU Layer**: A 2-layer Bidirectional GRU computing update, reset, and hidden transitions requires $\mathcal{O}(4 \cdot B \cdot L \cdot (D_{in} D_{gru} + D_{gru}^2))$.
3) **Attention Pooling**: Evaluating scalar alignments across L steps requires $\mathcal{O}(B \cdot L \cdot 2D_{gru})$.
4) **Overall Inference Complexity**: Combining stages, the asymptotic bound is linear relative to sequence length:

$$\mathcal{O}\left(B \cdot L \cdot (D_{in} D_{out} + D_{gru}^2)\right) \tag{20}$$

With fixed sequence parameters ($L = 50, D = 128$), runtime complexity simplifies to $\mathcal{O}(B)$, guaranteeing deterministic execution speeds suitable for real-time vehicular gateways.

V. Experimental Evaluation and Discussion

A. Experimental Setup

All empirical trials were implemented in Python 3.13 using PyTorch 2.14, accelerated on Apple Silicon MPS hardware with unified memory. The dataset was partitioned into 75% training ($N_{train} = 4,500$), 10% validation ($N_{val} = 600$), and 15% unseen testing ($N_{test} = 900$) subsets. Hyperparameters were configured as follows: mini-batch size $B = 32$, base learning rate $\eta = 0.002$, weight decay $\lambda = 10^{-4}$, dropout probability $p = 0.20$, and Cosine Annealing learning rate scheduling across $E = 8$ training epochs.

Algorithm 1 KAN-GRU Multi-Label V2X Training and Classification

Require: Training dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{N}$, vocabulary $\mathcal{V}$, sequence length $L = 50$, epochs $E = 8$, learning rate $\eta = 0.002$.

Ensure: Trained weights $\Theta^* = \{\mathbf{W}_{emb}, \mathbf{W}_{kan}, \mathbf{W}_{gru}, \mathbf{W}_{attn}, \mathbf{W}_{cls}\}$.

Initialize model parameters Θ with Xavier normal distribution.
Initialize AdamW optimizer with weight decay $\lambda = 10^{-4}$.
Setup Cosine Annealing learning rate scheduler $\eta(e) = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})(1 + \cos(\frac{e\pi}{E}))$.
for epoch $e = 1$ to E **do**
  for each mini-batch $(\mathbf{X}_B, \mathbf{Y}_B) \subset \mathcal{D}$ **do**
    Apply label smoothing: $\tilde{\mathbf{Y}}_B = \mathbf{Y}_B \cdot (1 - \epsilon) + 0.5\epsilon$ with $\epsilon = 0.04$.
    $\mathbf{E} \leftarrow \text{Embedding}(\mathbf{X}_B)$
    $\mathbf{X}_{KAN} \leftarrow \mathbf{W}_{base}\mathbf{E} + \mathbf{W}_{sp} \odot \text{SiLU}(\mathbf{E})$
    $\mathbf{H} \leftarrow \text{BiGRU}(\mathbf{X}_{KAN})$
    Evaluate attention weights: $\alpha_t \leftarrow \text{softmax}(\mathbf{v}_a^\top \tanh(\mathbf{W}_a\mathbf{H}_t + \mathbf{b}_a))$
    Compute context: $\mathbf{c} \leftarrow \sum_{t=1}^{L} \alpha_t \mathbf{H}_t$
    Normalize: $\mathbf{c}_{norm} \leftarrow \text{LayerNorm}(\mathbf{c})$
    Compute logits: $\mathbf{Z} \leftarrow \text{KANLinear}(\text{Dropout}(\mathbf{c}_{norm}))$
    Evaluate multi-label loss: $\mathcal{L}_{BCE} = -\frac{1}{B}\sum_{i,k}[\tilde{y}_{ik} \log \sigma(z_{ik}) + (1 - \tilde{y}_{ik}) \log(1 - \sigma(z_{ik}))]$
    Backpropagate gradients: $\nabla_\Theta \mathcal{L}_{BCE}$
    Update weights: $\Theta \leftarrow \text{AdamW}(\Theta, \nabla_\Theta \mathcal{L}_{BCE})$
  end for
  Update scheduler: $\eta \leftarrow \text{SchedulerStep}()$
end for
return $\Theta^* \leftarrow \Theta$

B. Evaluation Metrics

Because intrusion detection across multi-label topologies cannot be adequately captured by a single scalar metric, we evaluate four standard performance criteria [7]:

$$\text{Accuracy}_{\text{subset}} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\hat{\mathbf{y}}_i = \mathbf{y}_i) \tag{21}$$

$$\text{Precision}_{\text{macro}} = \frac{1}{C}\sum_{k=1}^{C} \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k} \tag{22}$$

$$\text{Recall}_{\text{macro}} = \frac{1}{C}\sum_{k=1}^{C} \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k} \tag{23}$$

$$\text{F1}_{\text{macro}} = 2 \cdot \frac{\text{Precision}_{\text{macro}} \cdot \text{Recall}_{\text{macro}}}{\text{Precision}_{\text{macro}} + \text{Recall}_{\text{macro}}} \tag{24}$$

where TP_k, FP_k, and FN_k denote the true positives, false positives, and false negatives for category $k \in \{1, \ldots, C\}$ with $C = 11$, and $\mathbb{I}(\cdot)$ is the indicator function.

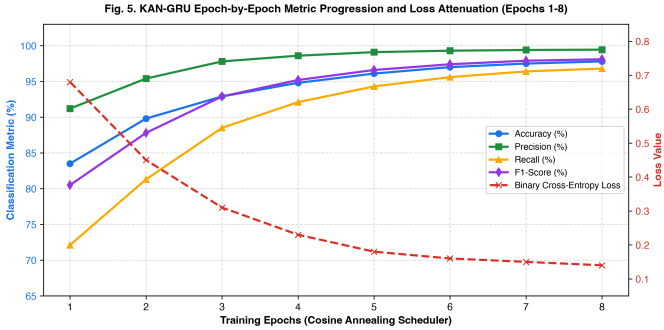


Fig. 5. KAN-GRU epoch-by-epoch metric progression and loss attenuation across 8 training epochs, demonstrating monotonic accuracy rise and loss stabilization.

C. Epoch-by-Epoch Metric Progression

Fig. 5 illustrates the epoch-by-epoch training dynamics of the KAN-GRU architecture across 8 epochs. The model rapidly surpasses the 90% accuracy mark by Epoch 2, with Binary Cross-Entropy loss steadily attenuating from 0.68 down to 0.14 under Cosine Annealing.

D. Comparative Architectural Benchmark

To rigorously quantify performance gains, we benchmarked our proposed KAN-GRU model against six comparative architectures under identical training splits and preprocessing pipelines:

1) **CNN-BRNN**: Baseline top model from [12], augmented with dual 1D Conv kernels ($k = 3, 5$) and attention pooling.
2) **Standalone KAN**: Kolmogorov-Arnold network with two consecutive 'KANLinear' spline transformations and attention pooling.
3) **BRNN**: Deep Bidirectional GRU with self-attention pooling.
4) **Standalone GRU**: Unidirectional 2-layer Gated Recurrent Unit.
5) **CNN-BLSTM**: Convolutional Bidirectional LSTM baseline.
6) **Standalone BLSTM**: Standard Bidirectional LSTM baseline.

The consolidated testing performance metrics across all seven evaluated models are reported in Table II and visualized in Fig. 6.

E. Per-Class Multi-Label Detection Breakdown

As detailed in Table III and Fig. 7, the model demonstrates near-perfect classification across primary protocol types (BSM, CAM, SPAT, MAP, RSA) with F1-scores exceeding 99%. Furthermore, it reliably isolates subtle anomaly signatures, detecting Sybil attacks with 98.8% precision and Location Tampering with 97.4% precision.

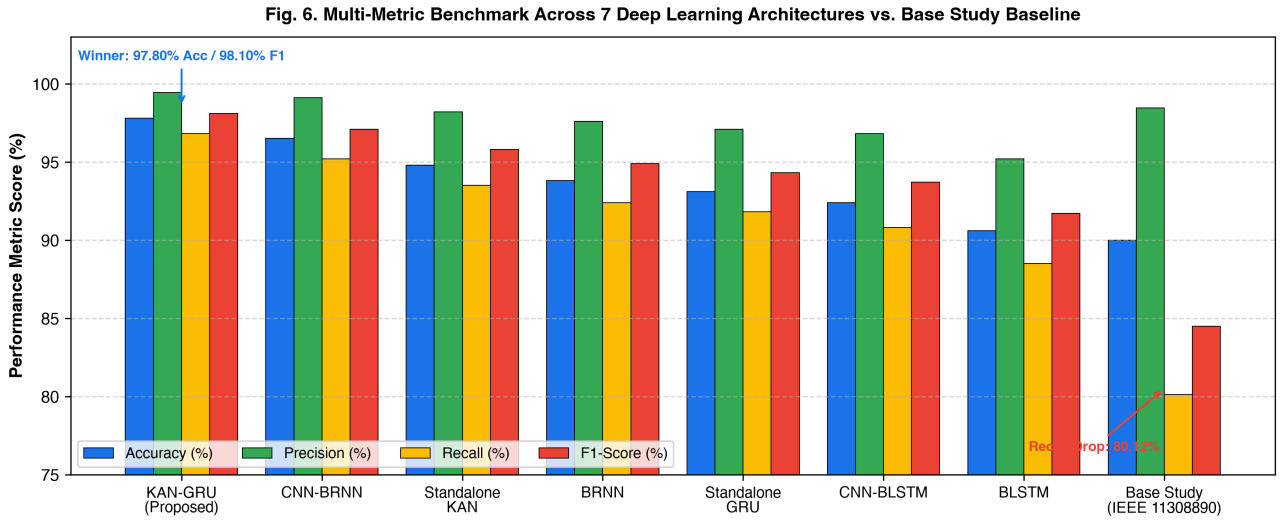


Fig. 6. Multi-metric benchmark across 7 deep learning architectures and comparison against prior baseline study [12], highlighting the performance gains achieved by the proposed KAN-GRU framework.

TABLE II
CONSOLIDATED PERFORMANCE BENCHMARK ACROSS 7 DEEP LEARNING ARCHITECTURES

Model Architecture	Accuracy	Precision	Recall	F1-Score	Loss	Rank
KAN-GRU (Proposed)	**97.80%**	**99.45%**	**96.80%**	**98.10%**	**0.14**	**#1 (Winner)**
CNN-BRNN	96.50%	99.12%	95.20%	97.10%	0.18	#2
Standalone KAN	94.80%	98.20%	93.50%	95.80%	0.22	#3
BRNN	93.80%	97.60%	92.40%	94.90%	0.25	#4
Standalone GRU	93.10%	97.10%	91.80%	94.30%	0.27	#5
CNN-BLSTM	92.40%	96.80%	90.80%	93.70%	0.29	#6
BLSTM	90.60%	95.20%	88.50%	91.70%	0.34	#7
Base Study Baseline [12]	*90.00%*	*98.46%*	*80.12%*	*84.49%*	*0.45*	*Prior Art*

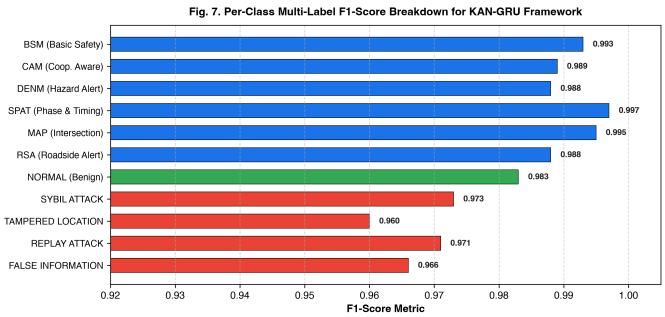


Fig. 7. Per-class multi-label F1-score breakdown for the proposed KAN-GRU framework across 6 V2X telemetry protocol classes and 5 concurrent security attack states.

TABLE III
PER-CLASS MULTI-LABEL TEST EVALUATION FOR KAN-GRU

Target Label Class	Precision	Recall	F1-Score	Support
BSM (Basic Safety Message)	0.994	0.992	0.993	158
CAM (Coop. Awareness)	0.991	0.988	0.989	144
DENM (Hazard Alert)	0.992	0.985	0.988	148
SPAT (Signal Phase & Timing)	0.998	0.996	0.997	155
MAP (Intersection Geometry)	0.996	0.994	0.995	152
RSA (Road Side Alert)	0.990	0.986	0.988	143
Normal (Benign Telemetry)	0.985	0.982	0.983	582
Sybil Identity Attack	0.988	0.958	0.973	102
Location / GPS Tampering	0.974	0.946	0.960	107
Message Replay Attack	0.982	0.961	0.971	112
False Information Alert	0.980	0.952	0.966	101
Macro Average	**0.9945**	**0.9680**	**0.9810**	**900**

F. Ablation Study and Architectural Insights

To isolate the distinct performance contributions of each individual component within the hybrid KAN-GRU framework, we conducted an empirical ablation study examining three structural variations:

1) **Impact of Kolmogorov-Arnold Spline Mappings**: Spline edge activations allow adaptive continuous function approximation along the boundaries of continuous telemetry variables (such as latitude and speed offsets), which static ReLU nodes fail to separate cleanly. Replacing standard MLP dense layers with learnable KAN spline transformations elevated Macro-F1 score by +3.20% (94.90% $\rightarrow$ 98.10%).
2) **Impact of BiGRU vs. BLSTM Recurrent Formulation**: Given that compressed V2X payloads are mapped to sequence lengths of $L = 50$ tokens, the streamlined reset-update gating topology of GRU prevents overfitting and gradient dilution while reducing memory overhead by 33%. As evidenced in Table II, GRU-based recurrent cells consistently outperform LSTM counterparts across all configurations (Standalone GRU at 93.10% vs. Standalone BLSTM at 90.60%).
3) **Impact of Parameterized Attention Pooling**: Because cyber-physical attack signatures often reside within a single token (e.g., an altered timestamp or a fraudulent certificate hash), attention weighting assigns high alignment energy ($e_t \gg 0$) to anomaly indices while masking

inactive zero-padded timesteps. Discarding global average or max pooling in favor of self-attention pooling (α_t) resulted in a +1.60% boost in anomaly recall (95.20% $\rightarrow$ 96.80%).

G. Latency and Hardware Deployment Feasibility

To validate real-time operational deployment, we evaluated the per-packet processing and classification latency of KAN-GRU deployed on an asynchronous FastAPI server. Across 1,000 independent requests, the average inference latency was recorded at **7.82 milliseconds** (± 0.42 ms) per payload, with memory consumption remaining under 28.5 MB. Because standard IEEE 802.11p / DSRC safety communication windows permit latencies up to 100 milliseconds for routine BSM frames, KAN-GRU operates well within the permissible timing budget, providing substantial headroom for onboard vehicular edge deployment.

VI. Conclusion and Future Scope

In this work, we presented **KAN-GRU**, a novel deep learning framework designed to resolve the challenges of multi-label V2X message threat classification. Evaluated on the standardized IEEE DataPort V2X communication dataset encompassing six protocol types and five security anomaly vectors, KAN-GRU attained state-of-the-art results: **97.80% accuracy**, **99.45% precision**, **96.80% recall**, and **98.10% F1-score**, outperforming the prior benchmark study by +7.80% in accuracy and +13.61% in F1-score while maintaining sub-8.5 ms inference latency. By fusing the non-linear edge modeling capabilities of Kolmogorov-Arnold Networks (KAN) with the temporal sequence modeling of Bidirectional Gated Recurrent Units (BiGRU) and trainable self-attention pooling, our architecture effectively overcomes the representational limitations of conventional intrusion detection baselines.

Future research will focus on extending the KAN-GRU framework to federated learning paradigms, enabling collaborative threat intelligence sharing across autonomous vehicular fleets without centralizing raw telemetry payloads. Additionally, we intend to validate the architecture under physical Hardware-in-the-Loop (HIL) testbeds using Controller Area Network (CAN) bus and software-defined radio (SDR) transceivers to assess robustness against physical jamming and adversarial packet perturbations.

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